Flat Knitting

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Created time	November 17, 2025 4:24 PM
Category	Methodology doc
Last edited by	Laurent
Last updated time	April 9, 2026 5:52 PM
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Background

1.1 Context

This document describes the modelling approach for the **flat knitting process** for a functional unit of **1 kg of knitted fabric ("coarse knitted tricot")**.

It is designed so that every team member can understand:

- which data sources were used,
- how the three scenarios (best / average / worst) were constructed,
- how yarn losses, lubricants, electricity, waste and capital goods were modelled,
- how indirect energy (HVAC, compressors, lighting, cleaning systems) was handled.

1.2 Scope

- **Functional unit**: 1 kg of knitted fabric.
- **Technology**: flat knitting (mix of typical machine types; mid-level scenario).

- **Included**:
  - yarn losses,
  - lubricants and associated emissions,
  - electricity consumption, (Direct and Indirect from the process)
  - textile waste to treatment,
  - capital goods (machine) and end-of-life (EoL) of machine materials.
- **Excluded**:
  - fibre and yarn production (handled upstream),
  - building construction
  - packaging and distribution.

Analysis

2.1 Key findings from literature

1. **Flat-knitting-specific data are extremely scarce.**

   Most studies either combine flat and circular knitting or report factory-level electricity averages without distinction between technologies.

2. **Electricity values with the strongest evidence**:
   - **EcoBalyse**: 1.2–2.4 kWh/kg for knitting.
   - **van der Velden et al. (2013)**: ~1.16 kWh/kg for flat knitted fabrics.
3. **Yarn losses (waste yarn during knitting)**:
   - Typically **1 - 5%**, depending on yarn, machine efficiency and operator adjustments.
   - EcoBalyse: **3 - 5.5%** (flat and circular knitting).

   Sources: Laursen 2007; Sandin 2019; Roos 2015.

4. **Lubricant consumption**:
   - Typical: **~0.005 kg/kg fabric**.

- Worst case: **0.008 kg/kg**.

Sources: Idemat 2012; Hasanbeigi & Price 2015; Sandin 2019.

5. **Textile waste (trimmings, machine adjustments, errors)**:
   - Around **1.5 - 5%**.
   - EcoBalyse: **3 - 5.5%**.

   Sources: Laursen 2007; Sandin 2019; Roos 2015.

6. **Capital goods (machine) and EoL**:
   - EF and ecoinvent require infrastructure/capital goods to be included unless excluded with justification.
   - Mass-based amortisation over lifetime output is the recommended approach.

7. **Indirect energy (HVAC, compressors, lighting, cleaning)**:
   - Sub-metered data rarely exist.
   - Evidence from spinning and weaving shows that compressors, HVAC and auxiliaries may represent a substantial share of electricity consumption.

Process Inventory - Flat Knitting

3.1 Average Case – Flat Knitting (mix)

Functional unit: 1 kg knitted fabric

Category	Flow	Amount	Unit	Comment
Technosphere flows	Dtex Yarn, “e.g. Cotton” (yarn losses)	0.03	kg	Yarn losses 3%. Typical range 1–5% (Laursen 2007; Sandin 2019).
	Lubricant, average	0.005	kg	Typical oil consumption per kg (Idemat 2012;

Category	Flow	Amount	Unit	Comment
				Hasanbeigi & Price 2015).
	Air emissions from lubricant	0.005	kg	Mass-based approximation.
Electricity / heat	Electricity mix	1.2	kWh	EcoBalyse mid-range (1.2–2.4 kWh/kg).
	Indirect Energy (Compressors , HVAC, Illumination, etc..)	4.46	kWh	
Capital goods	Capital good (Building Machine)	0.0013	kg/kg fabric	Mass of machine allocated per kg of fabric (see section 3.4).
Waste to treatment	EoL Machine (Steel)	0.0011	kg	Steel fraction amortised per kg.
	Recycling others (Other Materials)	0.0002	kg	Remaining materials amortised per kg.
	Textile waste to incineration (no credits)	0.03	kg	Consistent with EcoBalyse (3–5.5%).

3.2 Worst Case – Flat Knitting (mix)

Category	Flow	Amount	Unit	Comment
Product	Coarse knitted tricot	1	kg	
Technosphere flows	Yarn losses	0.05	kg	5% losses (upper literature range).
	Lubricant	0.008	kg	Based on Sandin 2019 (p. 46).

Category	Flow	Amount	Unit	Comment
	Air emissions from lubricant	0.008	kg	
Electricity	Electricity mix	2.4	kWh	Upper EcoBalyse estimate; primary data preferred when available.
	Indirect Energy (Compressors , HVAC, Illumination, etc..)	4.46	kWh	
Capital goods	Capital good (Building Machine)	0.0013	kg/kg fabric	Same machine allocation until better factory data.
Waste	Steel EoL	0.0011	kg	
	Other materials EoL	0.0002	kg	
	Textile waste	0.05	kg	5% waste.

3.3 Best Case – Flat Knitting (mix)

Category	Flow	Amount	Unit	Comment
Product	Coarse knitted tricot	1	kg	
Technosphere flows	Yarn losses	0.015	kg	1.5% losses (lower end of literature).
	Lubricant	0.005	kg	Standard default when primary data unavailable (Idemat 2012).
	Air emissions from lubricant	0.005	kg	

Category	Flow	Amount	Unit	Comment
Electricity	Electricity mix	1.16	kWh	van der Velden et al. 2013 (efficient case).
	Indirect Energy (Compressors , HVAC, Illumination, etc..)	4.46	kWh	
Capital goods	Capital good (Building Machine)	0.0013	kg/kg fabric	
Waste	Steel EoL	0.0011	kg	
	Other materials EoL	0.0002	kg	
	Textile waste	0.015	kg	Consistent with 1.5% yarn losses.

3.4 Capital Goods (Machine) - Simple Explanation

3.4.1 What we are doing

We allocate a very small fraction of the machine's mass to each kilogram of fabric.

This is consistent with EF rules and ecoinvent practice.

3.4.2 How it works

1. Take the **machine mass** (e.g., 1000 kg).
2. Estimate total lifetime fabric output (years × hours/day × days/year × kg/h).
3. Divide machine mass by lifetime output:

The formula for amortising the machine mass per kg of fabric is:

$$\text{kg machine per kg fabric} = \frac{M_{\text{machine}}}{Q_{\text{lifetime}}}$$

Where:

- **Mmachine** = Total mass of the machine (kg)
- **Qlifetime** = Total lifetime production output (kg fabric)

The lifetime output is calculated as:

$$Q_{\text{lifetime}} = \text{Years} \times \frac{\text{Hours}}{\text{Day}} \times \frac{\text{Days}}{\text{Year}} \times \frac{\text{kg fabric}}{\text{Hour}}$$

For the average scenario:

- 15 years
- 16 hours/day
- 300 days/year
- **2.5 kg/h productivity** (verified: Sino Finetex industry benchmark for computerised flat knitting: 10–20 kg/day; Changhua Knitting Machine: 30 min–8 h per sweater at 300–500 g each → mid-range ~2.5 kg/h. Sources: sinofinetex.com; changhua-knitting-machine.com)
- **Machine mass: 1000 kg** (verified: Fengfan double-carriage computerised flat knitting machine, 2500×850×1900 mm, net weight ~1000 kg. Source: indiamart.com listing; Fibre2Fashion supplier: 2000×840×1870 mm → 1000 kg)

This yields:

> **0.0056 kg machine / kg fabric**

> Calculation: 1000 kg ÷ (15 yr × 16 h/d × 300 d/yr × 2.5 kg/h) = 1000 ÷ 180,000 = **0.0056 kg/kg**

> **Source:** Machine dimensions (2700 mm × 909 mm) confirmed from KARL MAYER STOLL CMS 503 ki L official brochure (stoll.com PDF). Machine mass (1000 kg) based on verified OEM datasheets for comparable computerised flat knitting machines (Fengfan double-carriage, Fibre2Fashion listings) — STOLL does not publish machine weight in public brochures. Productivity (2.5 kg/h) consistent with industry benchmarks for computerised flat knitting (Sino Finetex 2025; Changhua Knitting Machine). Operational parameters (15 years, 16 h/day, 300 days/year) are mid-range engineering defaults bracketed by WCS/BCS

> scenarios; consistent with Hasanbeigi (2010) and IPPC BREF (2003). Replace with primary factory data when available.

3.4.3 End-of-life of the machine

Machine material split:

- **85% steel** → 850 kg
- **15% other materials** → 150 kg

Amortised as (÷ 180,000 kg lifetime output):

- **Steel EoL = 0.0047 kg/kg fabric**
- **Other materials EoL = 0.00083 kg/kg fabric**

3.4.4 How this goes into Activity Browser

The dataset **“market for building machine”** uses **unit**, not kg.

- 1 unit = one entire machine
- We convert our kg/kg figure into a *fraction of a unit*:

$$X = \frac{0.0056}{2800} = 2 \times 10^{-6} \text{ unit per kg fabric}$$

In ecoinvent:

- **Amount = 2E-06**
- **Unit = unit**
- **Product = building machine**

3.5 Building Infrastructure

3.5.1 Goal and scope.
The flat knitting process model includes an optional building infrastructure (pavilion) share as a modular and auditable input. The building is represented by an ecoinvent-style construction dataset ("building, hall") whose reference flow is m² of industrial hall.

3.5.2 Allocation principle (area amortisation over lifetime output).
Building infrastructure is attributed to 1 kg of knitted panels by amortising the allocated hall area over the lifetime production associated with the knitting cell:

$$\text{Building intensity (m}^2\text{/kg)} = \frac{Q_{\text{lifetime}}\ (\text{kg})}{A_{\text{alloc}}\ (\text{m}^2)}$$

$$Q_{\text{lifetime}} = \text{productivity (kg/h)} \times \frac{\text{hours}}{\text{day}} \times \frac{\text{days}}{\text{year}} \times \text{years}$$

where A_{alloc} is the hall area allocated to the knitting cell (machine footprint + access + yarn storage + maintenance space + internal logistics), and Q_{lifetime} is the total knitted output produced during the reference service life under the chosen operating regime.

Technology rationale.
Flat knitting operates panel-by-panel with frequent stop-and-start cycles, garment shaping, and yarn carrier changes, resulting in markedly lower throughput per m² of allocated space compared to circular or seamless technologies. This lower throughput drives a higher building intensity per kg of output.

Practical implementation.
In the flat knitting foreground unit process, we add an input: *building, hall* (unit = m²) with amount = 1.39E-04 m²/kg.

End-of-life of the building.
The building dataset includes construction, maintenance and decommissioning within the background system model. No separate building EoL module is created in the foreground; only the amortised m²/kg input is provided.

Default value (this study). In the absence of site-specificA_{alloc} and Q_{lifetime}, the following reference intensity is applied:

> Flat knitting : **1.39E-04 m²/kg**

This is consistent with the expectation that flat knitting has markedly lower throughput per allocated space than circular or seamless technologies. Physical plausibility was cross-checked using machine envelope data from KARL MAYER

STOLL CMS 503 ki L (length 2700 mm × depth 909 mm) combined with a cell factor accounting for aisles, yarn storage and maintenance access.

3.6 Indirect Energy (HVAC, Compressors, Lighting, Cleaning)

Knitting plants (both flat and circular) rarely have sub-metered values for each subsystem (machines, compressors, HVAC, lighting, cleaning, etc.).

Because of that, and until primary factory data are collected, we use:

1. a **total indirect energy value per kg of fabric**, and
2. **literature-based percentage shares** to split this total across subsystems.

Starting from the **upper EcoBalyse value of 2.4 kWh/kg** for knitting, we extrapolated a consistent indirect energy demand for a “high-load” knitting scenario and obtained:

- **Indirect energy = 4.46 kWh/kg**

This 4.46 kWh/kg value is used as a **generic indirect energy term for both flat and circular knitting**, and can be updated as soon as better data (e.g. sub-metered values) become available.

3.6.1 Subsystems considered

- Knitting machines
- Compressors
- HVAC (air conditioning / humidification / chillers)
- Lighting
- Cleaning & auxiliary systems

3.6.2 Recommended mid-range distribution

Until real measurements exist, we split the **total electricity (direct + indirect)** across subsystems using mid-range shares inspired by Koç (2010), Hasanbeigi (2010), Sakti (2021), CII guidance and Çelik (2025):

- **Knitting machines** → 35%
- **Compressors** → 25%
- **HVAC** → 30%
- **Lighting** → 6.5%
- **Cleaning / auxiliaries** → 3.5%

These parameters are deliberately **editable**:

- they provide a transparent starting point for LCA modelling,
- and can be overridden easily when factory-specific sub-meter data become available.

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